Plotly INTORDUCTION

Python Plotly Library is an open-source library that can be used for data visualization and understanding data simply and easily.

Plotly supports various types of plots like line charts, scatter plots, histograms, cox plots, etc.

Installation

Plotly does not come built-in with Python.

```
pip install plotly
```

Package Structure of Plotly

There are three main modules in Plotly. They are:

- plotly.plotly
- plotly.graph.objects
- plotly.tools

the visualizations are interactive unlike Seaborn and Matplotlib;

it's quite straightforward to generate complicated visuals using Plotly's high-level Express API;

We can interact with the Plotly API in two ways;

Plotly Graph Objects lower-level API

Plotly Express high-level API

plotly.plotly acts as the interface between the local machine and Plotly. It contains functions that require a response from Plotly's server.

plotly.graph_objects module contains the objects (Figure, layout, data, and the definition of the plots like scatter plot, line chart) that are responsible for creating the plots.

The Figure can be represented either as dict or instances of plotly.graph_objects.

Note: plotly.express module can create the entire Figure at once. It uses the graph_objects internally and returns the graph_objects.

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Line Chart

Line plot in Plotly is much accessible and illustrious annexation to plotly which manage a variety of types of data and assemble easy-to-style statistic.
Line chart Displays a series of numerical data as points which are connected by lines.
It visualizes to show two data trends.
The main productive feature is it can display thousands of data points without scrolling.
It can be created using the line() method of plotly.express class.

We have created a simple line chart by passing the x, y coordinates of the points to be plotted.

```
import plotly.express as px

a=[1,2,3]
b=[1,2,3]
fig=px.line(x=a,y=b)
fig.show()
```

Bar Chart

A bar chart is a pictorial representation of data that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent.

In other words, it is the pictorial representation of dataset.

These data sets contain the numerical values of variables that represent the length or height.

```
import plotly.express as px

# using the iris dataset
df = px.data.iris()
df
fig = px.bar(df, x="sepal_width", y="sepal_length")

# showing the plot
fig.show()
```

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Histograms

A histogram contains a rectangular area to display the statistical information which is proportional to the frequency of a variable and its width in successive numerical intervals.

A graphical representation that manages a group of data points into different specified ranges.

It has a special feature that shows no gaps between the bars and similar to a vertical bar graph.

```
import plotly.express as px

df = px.data.iris()
# plotting the histogram
fig = px.histogram(df, x="sepal_length", y="petal_width")

# showing the plot
fig.show()
```

Histograms in Plotly using graph_objects class

```
import plotly.graph_objects as go
import plotly.express as px
df = px.data.iris()
fig = go.Figure()
fig.add_trace(go.Histogram(x=df['sepal_width']))
fig.add_trace(go.Histogram(x=df['species_id']))
fig.update_layout(barmode='stack')
fig.show()
```

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Scatter Plot

A **scatter plot** is a diagram where each value is represented by the dot graph. Scatter plot needs arrays for the same length, one for the value of x-axis and other value for the y-axis.
Each data is represented as a dot point, whose location is given by x and y columns.
It can be created using the scatter() method of plotly.express

```
import plotly.express as px

#  iris dataset
df = px.data.iris()

fig = px.scatter(df, x="sepal_width", y="sepal_length",
                 color="species")
fig.show()
```

Changing Size

The size of the dots of the scatter plot can be changed using the size argument of the scatter() method.

Using

`size='petal_length'`

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Bubble Chart

The bubble chart in Plotly is created using the scatter plot.
It can be created using the **scatter()** method of plotly.express.
A bubble chart is a data visualization which helps to displays multiple circles (bubbles) in a two-dimensional plot as same in scatter plot.
A bubble chart is primarily used to depict and show relationships between numeric variables.

```
import plotly.express as px

df = px.data.iris()

fig = px.scatter(df, x="sepal_width", y="sepal_length",
                 color="species",
                 size='petal_length',)

fig.show()
```

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Pie Charts

A pie chart is a circular statistical graphic, which is divided into slices to illustrate numerical proportions.

It depicts a special chart that uses "pie slices", where each sector shows the relative sizes of data.

A circular chart cuts in a form of radii into segments describing relative frequencies or magnitude also known as circle graph.

```
import plotly.express as px

# using the tips dataset
df = px.data.tips()

# plotting the pie chart
fig = px.pie(df, values="total_bill", names="day")

# showing the plot
fig.show()
```

Box Plots

A Box Plot is also known as Whisker plot is created to display the summary of the set of data values having properties like minimum, first quartile, median, third quartile and maximum.

In the box plot, a box is created from the first quartile to the third quartile, a vertical line is also there which goes through the box at the median.

Here x-axis denotes the data to be plotted while the y-axis shows the frequency distribution.

```
import plotly.express as px

# using the tips dataset
df = px.data.tips()

# plotting the box chart
fig = px.box(df, x="day", y="total_bill")

# showing the plot
fig.show()
```

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Violin plots

Violin Plot is a method to visualize the distribution of numerical data of different variables.

It is similar to Box Plot but with a rotated plot on each side, giving more information about the density estimate on the y-axis.

The density is mirrored and flipped over and the resulting shape is filled in, creating an image resembling a violin.

The advantage of a violin plot is that it can show nuances in the distribution that aren't perceptible in a boxplot.

On the other hand, the boxplot more clearly shows the outliers in the data.

```
import plotly.express as px

# using the tips dataset
df = px.data.tips()

# plotting the violin chart
fig = px.violin(df, x="day", y="total_bill")

# showing the plot
fig.show()
```

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3D Scatter Plot Plotly

3D Scatter Plot can plot two-dimensional graphics that can be enhanced by mapping up to three additional variables while using the semantics of hue, size, and style parameters.

All the parameter control visual semantic which are used to identify the different subsets.

Using redundant semantics can be helpful for making graphics more accessible. It can be created using the scatter_3d function of plotly.express class.

```
import plotly.express as px

# Data to be plotted
df = px.data.iris()

# Plotting the figure
fig = px.scatter_3d(df, x = 'sepal_width',y = 'sepal_length',z =
'petal_width',color = 'species')

fig.show()
```